

Object Detection Extension

<https://github.com/aiyshwary/Object-Detection-Extension-in-Rapid-Miner>

[Python](#)

[PyTorch](#)

[ONNX Runtime](#)

[RapidMiner](#)

[Object Detection](#)

OVERVIEW

A RapidMiner Studio extension that brings transfer learning-based object detection into a no-code environment, supporting fine-tuning and inference with Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet models.

IMPACT

Developed two RapidMiner operators for object detection — `FineTuneObjectDetectionModel` and `ObjectDetectionModelInference` — enabling users to train and deploy detection models within a visual workflow without writing code.

TECHNICAL WALKTHROUGH

Object Detection Extension

Problem Statement

Different object detection use cases require different trade-offs between accuracy, speed, and computational cost. A single model cannot satisfy all scenarios.

Approach

Developed a unified object detection extension using PyTorch, supporting

multiple architectures

1. Faster R-CNN

2. FCOS

3. SSD

4. SSD Lite

5. RetinaNet

6. YOLO (extended work)

7. Training

8. Dataset annotated using XML bounding boxes

9. Transfer learning using COCO pre-trained weights

Fully configurable training pipeline

1. Model selection

2. Batch size

3. Epochs

4. Learning rate

5. Momentum

6. Weight decay

Evaluation metrics

1. IoU (Intersection over Union)

2. Precision

3. Recall (per class)

4. Inference

5. Users can choose any trained or pre-trained model

Output includes

1. Image with bounding boxes, labels, and confidence scores

2. Tabular output with detected objects

3. Results also generated in base64 format for easy deployment and

4. integration

5. Key Observations

6. Faster R-CNN performed best for accuracy

7. SSD and SSD Lite were faster but had lower recall

8. RetinaNet handled class imbalance well

9. Model choice depends on application requirements

KEY DETAILS

1. `FineTuneObjectDetectionModel` accepts image and annotation directories (YOLO/Pascal VOC) and outputs model weights and class list CSV.

2. `ObjectDetectionModelInference` runs predictions on new images and returns bounding boxes and class labels as a `DataFrame`.

3. Supports five architectures: Faster R-CNN, FCOS, SSD, SSDLite, and RetinaNet for varying speed-accuracy trade-offs.

4. Configurable for CPU-only and GPU (CUDA cu118) environments; installed as a JAR extension in RapidMiner Studio.

5. Enables domain-specific object detection with minimal ML expertise using a drag-and-drop process designer.